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# Enabling Complexity: A Systematic Literature Review on Task Allocation, Communication, Interaction, and Augmentation in Human-AI Teams

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## Abstract

As artificial intelligence (AI) systems increasingly act as collaborators rather than tools, Human-AI Teams (HATs) are emerging in domains that demand complex, adaptive decision-making. This study investigates how four key enablers—task allocation, communication, interaction, and cognitive augmentation—support Human-AI collaboration in complexity-rich environments. A systematic literature review (SLR) of 50 peer-reviewed studies was conducted, with 39 rated as high-quality, supported by a custom Human-AI review environment combining natural language processing (NLP) and large language models (LLMs). The findings reveal recurring patterns, interdependencies, and research gaps across domains such as healthcare, logistics, and strategy. While each enabler has been examined independently, their combined role remains underexplored. Results highlight that structured task delegation, explainable and contextual communication, and alignment between AI augmentation and human intent significantly enhance team performance and trust. Nonetheless, issues such as user overreliance, cognitive misalignment, and transparency—particularly in the context of emerging Generative AI (GenAI) tools—remain key challenges for effective collaboration. This review contributes a synthesized understanding of effective hybrid teaming and demonstrates how Human-AI collaboration can improve not only task performance but also the research process itself. The study advances the field of Human-Centered AI and supports the vision of Industry 5.0 through actionable insights into hybrid team design.

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## 1. Introduction

As AI evolves, organizations increasingly integrate AI not merely as an automation tool but as a cooperative partner that enhances human decision-making. This transformation fosters Human-AI Collaboration (HAC), where AI systems—ranging from recommendation engines to large language models—work alongside humans by supporting

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cognitive tasks, facilitating workflow coordination, and ensuring adaptive teamwork while maintaining human oversight [1, 2]. In these hybrid teams, effectiveness hinges on how collaboration is structured across technical and human dimensions. While traditional enablers such as **task allocation** and **communication** remain essential, the integration of AI introduces additional dependencies on **interaction design** and **cognitive augmentation**. These support mutual understanding, coordination, and trust—especially when humans and AI agents differ in capabilities, context awareness, or autonomy [3]. Yet, the *combined influence of these four enablers*—under conditions of uncertainty, dynamism, and interdependence—remains underexplored. Much of the current literature examines these mechanisms in isolation or within narrow domains such as healthcare or finance, limiting our ability to generalize across complex environments. To address this, a **SLR** was conducted to examine:

*How do task allocation, communication, interaction, and cognitive augmentation influence Human-AI team performance in complex task environments?*

This review targeted 50 peer-reviewed empirical and conceptual studies, drawing on fields such as human-computer interaction, AI-supported decision-making, and collaborative systems. It aims to consolidate existing knowledge on Human-AI team collaboration, with a particular focus on task allocation, communication, interaction, and augmentation. By synthesizing findings across domains and methodologies, it seeks to uncover prevailing patterns, identify research gaps, and inform future research and practical implementation. In doing so, this work contributes to ongoing discussions in **Human-Centered AI (HCAI)** and the evolution towards **Industry 5.0**, which envision a future where humans and AI operate not in parallel—but as *adaptive, synergistic partners* in complex socio-technical systems [4, 5].

## 2. Background

### 2.1. Human-AI Teams in Complex Collaboration

The transition from AI as a mere tool to an active collaborative agent fosters hybrid intelligence by integrating the complementary strengths of both human and AI teammates, enhancing adaptability, efficiency, and collective problem-solving. [6, 7, 8]. As AI systems adopt natural language and predictive capabilities, the line between tool and teammate becomes increasingly blurred—raising challenges around shared mental models, transparency, and agency negotiation [9]. Addressing these challenges requires interaction frameworks that support adaptive teaming, aligning with the Industry 5.0 vision of human-centric, sustainable AI in interdependent systems [10, 11].

### 2.2. Complexity in Hybrid Teams

HATs increasingly operate within complex systems characterized by non-linearity, emergence, and unpredictability [12, 13]. In such environments, AI contributes speed, scalability, and pattern recognition, while humans provide contextual reasoning, ethics, and adaptability [14, 15]. Effective teaming demands more than technical interoperability—it requires shared goals, cognitive alignment, and coordination mechanisms like joint attention and situational awareness [16]. Viewed as distributed cognitive systems, the success of HATs depends on interaction patterns, not just the competence of individual agents [17].

### 2.3. Enablers of Human-AI Collaboration

Effective Human-AI Collaboration (HAC) in complex environments is supported by four interdependent enablers: **task allocation**, **communication**, **interaction**, and **cognitive augmentation**. These enablers are not arbitrarily selected—they are grounded in decades of research in team science and supported by well-established theories such as *Shared Mental Models* (SMMs) and *Transactive Memory Systems* (TMSs). These frameworks emphasize that effective collaboration depends on mutual understanding of goals, roles, and knowledge distribution within a team.

**Task allocation** leverages TMS structures to assign responsibilities based on who possesses relevant expertise, ensuring coordination and reducing redundancy [18]. **Communication** plays a critical role in establishing and maintaining shared mental models across team members, enabling alignment of understanding and intent [19]. **Interaction**

mechanisms—such as turn-taking, feedback loops, and routine coordination—are essential for dynamically constructing and updating these cognitive structures during collaborative work [18]. Finally, **augmentation** enhances the team's collective cognitive capabilities by introducing external representations (e.g., AI tools, interfaces, visualizations) that scaffold decision-making, shared understanding, and problem-solving [20].

These enablers, deeply rooted in human team science, translate meaningfully into Human-AI contexts. Here, AI systems do not act as passive tools but as active cognitive collaborators within distributed socio-technical systems, shaping and being shaped by human interaction, task structures, and shared understanding.

*Task Allocation.* Involves distributing responsibilities based on context, uncertainty, system confidence, and human trust. Unlike human teams, HATs require explicit, adaptive delegation strategies [21, 22].

*Communication.* Refers to verbal, visual, and symbolic exchanges that enable intention alignment. Given asymmetries in capabilities and understanding, explainable AI and structured feedback are essential [23, 24].

*Interaction.* Shapes how humans and AI coordinate timing, turn-taking, and feedback. Well-designed interaction supports shared understanding and recovery from errors [25, 26].

*Augmentation.* Describes how AI extends human insight, foresight, and decision-making. Alignment with human strategies enhances performance, while misalignment risks overreliance or confusion [27, 28].

While extensively studied in isolation, these enablers are rarely synthesized. This review addresses that gap by exploring their combined role in supporting effective Human-AI collaboration under complexity.

### 3. Review Design and Methodology

This study follows a structured SLR to answer the research question, **how task allocation, communication, interaction, and cognitive augmentation do influence Human-AI team performance in complex task environments**. The protocol draws on PRISMA, PICO, and SPIDER frameworks to ensure rigor and reproducibility. A Human-AI teaming environment supported the review process, combining NLP, retrieval-augmented generation (RAG), and LLMs such as GPT-4.5, Mistral, and Phi. A Streamlit interface facilitated transparency and task distribution based on confidence, capability, and complexity—reflecting the very hybrid teaming principles under investigation.

#### 3.1. Protocol Overview

The review proceeded through seven phases:

- **Planning:** Research questions and eligibility criteria were developed using PICO and SPIDER heuristics.
- **Search:** Boolean queries were iteratively constructed and executed across IEEE Xplore, ACM DL, Springer-Link, ScienceDirect, and Web of Science. LLMs assisted with query refinement.
- **Screening:** Title/abstract screening was semi-automated using RAG-supported LLMs; full-text inclusion was manually confirmed. PRISMA guidelines were followed throughout.
- **Extraction:** Structured forms captured metadata, methods, findings, and research question (RQ) alignment. LLM-generated entries were manually validated.
- **Quality Assessment:** Studies were rated across six dimensions (e.g., relevance, rigor, bias) using a rubric based on Kitchenham.
- **Synthesis:** Studies were clustered by enabler theme and methodology; high-confidence patterns were synthesized across RQs.
- **Reporting:** PRISMA flow diagrams, tables, and visual summaries communicated results.

#### 3.2. Inclusion and Exclusion Criteria

Criteria were established during the planning phase and integrated into the screening system, ensuring that ambiguous cases are flagged for human review in Table 1.

### 3.3. Search Strategy

Search strings were developed around the four enabler themes and Human-AI teaming. Boolean structures and synonyms were tested across the five databases. Query design was refined using metadata from OpenAlex and Crossref. Full queries are available in Appendix A.

### 3.4. Screening and Study Selection

Screening followed PRISMA 2020 and included three steps in Figure 1:

*Deduplication and Filtering.* From 1,449 retrieved records, 906 duplicates were removed. Filtering by date, language, and type reduced the set to 152.

*Title and Abstract Screening.* LLMs in a RAG pipeline evaluated relevance to Human-AI teaming and enabler dimensions. Ambiguities were flagged for manual review, yielding 52 studies for full-text screening.

*Full-Text Review.* 2 studies were excluded due to conceptual misalignment, leaving **50 studies** for synthesis.

### 3.5. Data Extraction

A structured form captured bibliographic and content-specific variables, following Kitchenham et al. [29] and PRISMA guidelines.

*Fields Included:*

- **Metadata:** Authors, year, DOI, venue
- **Study Design:** Experimental, qualitative, mixed
- **Domain:** Application context (e.g., healthcare, defense)
- **Enabler Focus:** Task allocation, communication, interaction, augmentation
- **Findings:** Key results and indicators
- **Relevance:** Mapping to RQ1–RQ4
- **Additional Notes:** Contributions, limitations

*Human-AI Workflow.* LLMs (GPT-4.5, Mistral, Phi) processed semantically chunked PDFs and generated field entries. These were verified using a Streamlit interface enabling traceable, side-by-side review.

*Output Format.* Data were exported in Markdown and CSV, supporting synthesis and visualization.

Table 1. Inclusion and Exclusion Criteria

| Criterion        | Inclusion                         | Exclusion                         |
|------------------|-----------------------------------|-----------------------------------|
| Language         | English or reliably translated    | Non-English                       |
| Publication Date | 2015–2025                         | Pre-2015 (unless foundational)    |
| Field            | CS, AI, HCI, Engineering          | Non-technical social sciences     |
| Domain           | ML, NLP, LLMs, GenAI, AI services | Symbolic/classical AI             |
| Study Type       | Empirical, SLRs, case studies     | Editorials, opinions, theory-only |
| Relevance        | Focus on enablers in team context | Generic HCI/AI                    |
| Peer Review      | Peer-reviewed                     | Non-peer-reviewed                 |
| Grey Literature  | High-quality vetted reports       | Blogs, unvetted preprints         |
| Duplication      | Most comprehensive version        | Redundant entries                 |

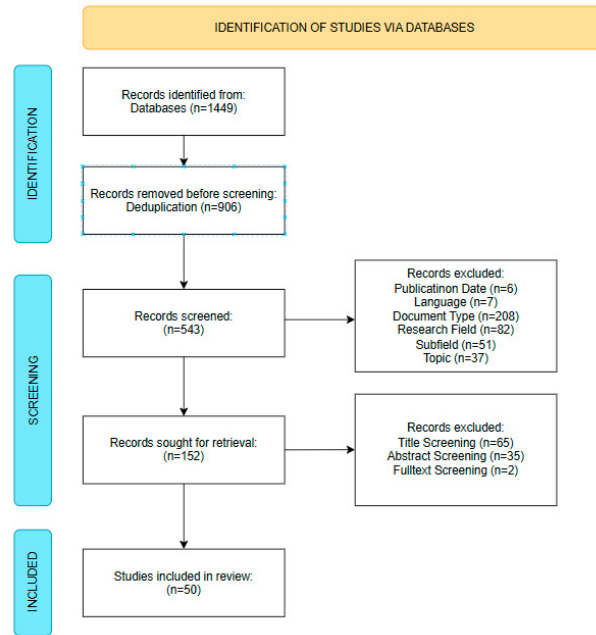


Fig. 1. PRISMA 2020 Flow Diagram for Study Selection

### 3.6. Quality Assessment

Each study was evaluated using a structured rubric aligned with quality assessment principles proposed by Kitchenham [29]. The rubric—tailored to the context of Human-AI teaming—covers six dimensions: relevance to the research questions, methodological rigor, sample adequacy, bias risk, theoretical contribution, and overall confidence. Table 2 details the scoring dimensions and scales used in the evaluation process.

Studies were categorized into three quality tiers based on these assessments. Table 3 shows the number of studies in each tier and their role in the synthesis process.

High-quality studies form the foundation of the synthesis in Section 4. Medium-quality papers are used for supporting observations, while low-quality studies are excluded from direct inference unless they offer unique insights.

Table 2. Quality Assessment Criteria and Scoring

| Criterion                | Description                                                          | Scoring             |
|--------------------------|----------------------------------------------------------------------|---------------------|
| Relevance to RQs         | Degree to which the study addresses one or more research questions.  | 1 (low) to 5 (high) |
| Methodological Rigor     | Appropriateness and transparency of research design and execution.   | 1 (low) to 5 (high) |
| Sample Adequacy          | Suitability of dataset or participant base for conclusions.          | 1 (low) to 5 (high) |
| Bias Risk / Limitations  | Presence and handling of bias or confounding factors.                | Low, Medium, High   |
| Theoretical Contribution | Contribution of conceptual or framework-based insights to the field. | Yes, Partial, No    |
| Overall Confidence       | Combined measure reflecting all criteria above.                      | High, Medium, Low   |

Table 3. Study Distribution by Quality Tier

| Quality Tier   | Number of Studies | Characteristics                                                            | Use in Synthesis                     |
|----------------|-------------------|----------------------------------------------------------------------------|--------------------------------------|
| High Quality   | 39                | Relevance and rigor $\geq 3$ , low/moderate bias, conceptual contribution. | Prioritized for core findings.       |
| Medium Quality | 9                 | One score below 3, or moderate limitations in method or scope.             | Supplementary; flagged in narrative. |
| Low Quality    | 2                 | Low relevance and/or rigor; high risk of bias.                             | Excluded or used only with caution.  |

### 3.7. Synthesis and Analysis

Following quality assessment, synthesis and analysis were conducted to consolidate findings across the four core enablers: *task allocation*, *communication*, *interaction*, and *cognitive augmentation*. This phase aimed to identify patterns, contradictions, and conceptual gaps relevant to Human-AI Team performance in complex tasks.

*Thematic Clustering.* Included studies were clustered into thematic categories based on their primary focus. The clustering process was supported by inductive coding and LLM-assisted classification using semantic similarity matching and keyword extraction. Each study could belong to multiple themes due to overlapping contributions.

- **Task Allocation (42 studies):** Studies addressing how responsibilities are shared or dynamically assigned between humans and AI.
- **Communication (40 studies):** Research exploring verbal, visual, or symbolic information exchange and transparency.
- **Interaction Design (38 studies):** Papers focused on system interface design, feedback mechanisms, and coordination protocols.
- **Cognitive Augmentation (24 studies):** Studies on how AI enhances or supports human cognition, judgment, or decision-making.

*Methodological Cross-Analysis.* To triangulate findings, the thematic clusters were cross-referenced with methodological types. This allowed for pattern analysis across different research designs:

- **Experimental (22 studies):** Quantitative studies testing the effect of enablers on team performance or user trust.
- **Qualitative (10 studies):** Interviews, case studies, or ethnographic work exploring user perception and team dynamics.
- **Mixed Methods (8 studies):** Studies combining statistical evidence with contextual analysis.
- **Conceptual/Theoretical (5 studies):** Frameworks or design principles for HATs.
- **Review/Systematic Review (5 studies):** Prior syntheses on related aspects such as explainable AI or trust.

*Confidence-Weighted Synthesis.* To prioritize evidence, synthesis focused on the 39 studies rated as high quality. Medium-quality studies (n=9) were used to support or qualify observations, while low-quality studies (n=2) were excluded from inference.

Synthesis was performed in the following stages:

1. **Open Coding:** Initial descriptive labels were applied to study findings based on extracted text and LLM support.
2. **Axial Coding:** Related codes were grouped into categories aligned with the four enablers.
3. **Narrative Integration:** Patterns were synthesized across categories to formulate higher-order insights.
4. **Gap Identification:** Underrepresented themes, contradictory findings, and limitations were documented.

The outcome of this synthesis directly informs the results section, where each research question is addressed using evidence-weighted thematic summaries. Tables and visuals are used to enhance traceability and present core findings, contradictions, and research gaps in a structured format.

## 4. Results

Findings are organized by the four guiding RQs and synthesized from 50 studies were included, with 39 rated as high-quality and forming the core synthesis, across diverse domains.

**RQ1: How is task allocation performed in Human-AI teams?** Task allocation is driven by *context-awareness*, *capability-matching*, and *conditional delegation*. Studies highlight the importance of **complementary expertise**, where humans manage ambiguity and ethics, while AI handles structured, data-heavy tasks. Adaptive models such as shared mental models and conditional autonomy enhance performance and reduce cognitive load.

### Key Findings:

- Conditional delegation based on system confidence improves outcomes [30, 8].
- Complementarity between human and AI strengths increases reliability [31].
- Clear role assignment fosters trust and reduces coordination overhead [32].

### RQ2: What communication strategies are used in Human-AI collaboration?

Communication in HATs emphasizes **explainability**, **bidirectionality**, and **multimodal feedback**. Explainable AI (XAI) improves user trust and task comprehension, especially under time pressure. Personalized, context-sensitive communication improves alignment and decision-making.

### Key Findings:

- Contextual explanations reduce cognitive friction and enhance accuracy [33, 11].
- Shared situational awareness supports high-stakes coordination [16].
- Multimodal outputs improve interpretability and reduce error rates [34].

### RQ3: How does AI augmentation influence decision-making in teams?

AI augmentation supports judgment through **prediction**, **pattern recognition**, and **insight generation**. It improves accuracy and reduces workload in data-rich environments. However, overreliance, deskilling, and opacity remain risks, especially when augmentation misaligns with user strategies.

### Key Findings:

- Augmentation improves decision speed and accuracy [35, 36].
- Misalignment can cause disengagement or overload [10, 34].
- Ethical concerns arise when AI overrides human agency [37, 38].

### RQ4: What role does GenAI play in Human-AI teams?

GenAI enhances **ideation**, **content generation**, and **early-stage design**. While effective for brainstorming and prototyping, it presents challenges in transparency, explainability, and bias. Most current work is conceptual, indicating a need for empirical validation.

### Key Findings:

- GenAI accelerates ideation and creative tasks [39, 15].
- Trust is limited by output opacity and non-determinism [6, 40].
- Empirical studies on GenAI in diverse team contexts are scarce.

**Summary.** Human-AI teams benefit from structured task delegation, contextual communication, and well-aligned cognitive augmentation. However, challenges around trust, transparency, and GenAI integration highlight key areas for future research.



## 5. Discussion

This review highlights the interdependent roles of **task allocation**, **communication**, **interaction**, and **augmentation** as key enablers of effective Human-AI Teams (HATs). Rather than functioning in isolation, these dimensions form a socio-technical coordination system that supports hybrid performance in complex environments. High-performing teams demonstrated structured delegation, transparent feedback, and context-aware AI support—suggesting that effectiveness arises not from replacing human judgment, but from extending it through alignment with AI capabilities. Dynamic task allocation and explainable, multimodal communication were particularly valuable in high-stakes or uncertain settings. Augmentation consistently improved accuracy and reduced cognitive load; however, several studies emphasized the risk of overreliance, underlining the importance of *calibrated trust*. Generative AI (GenAI) showed promise in ideation and content generation but also introduced challenges around traceability, bias, and ethical accountability.

The findings support Human-Centered AI (HCAI) by framing HATs as *distributed cognitive systems*, where collaboration effectiveness depends more on interaction quality than individual competencies. This aligns with theories of joint activity and cognitive engineering [7, 41], emphasizing trust calibration and fluid interaction design as essential for sustained teaming. The synthesis also supports the Industry 5.0 vision of *human-machine synergy*, offering practical guidance on designing AI systems that enhance—rather than displace—human agency.

From a methodological perspective, the literature was dominated by experimental and mixed-method studies, particularly in healthcare, emergency response, and collaborative design. Qualitative studies provided unique insights into user perception, emotion, and trust—highlighting the need for deeper interdisciplinary approaches in future work.

Despite rigorous procedures, this review has limitations. Domain-specific or grey literature may be underrepresented. The field of GenAI remains immature, with many contributions still conceptual. Thematic overlap across enablers complicated classification, and while LLMs supported the review process, manual oversight was necessary to mitigate hallucination risks.

Future research should prioritize longitudinal studies in real-world HAT contexts—particularly in engineering and safety-critical domains. Opportunities also exist to explore hybrid strategies for task allocation, communication, and augmentation in the development of complex artifacts. Further inquiry into trust calibration and ethical safeguards in GenAI is essential, as is the development of scalable frameworks that integrate LLMs and retrieval-augmented generation (RAG) systems to support complex knowledge work.

## 6. Conclusion

This study explored how HATs tackle complex tasks through four interdependent enablers: task allocation, communication, interaction, and cognitive augmentation. A systematic literature review of 50 peer-reviewed studies—39 of which were rated as high-quality—revealed recurring patterns, interdependencies, and key research gaps across various domains. Findings indicate that effective HATs rely on adaptive task allocation models, where responsibility shifts dynamically based on system confidence, human expertise, and situational demands. Conditional delegation and shared mental models enhance team performance by reducing ambiguity and improving response times. In practice, this is achieved through capability-aware systems that adjust roles in real-time, particularly in time-sensitive environments. Communication strategies in successful HATs emphasize explainability, bidirectionality, and multimodal feedback. Contextualized explanations and structured communication templates improve user understanding, reduce coordination errors, and foster mutual trust—especially when human and AI capabilities are asymmetric. Interaction design plays a central role in enabling coordinated action. Turn-taking, feedback loops, and interface design influence not only efficiency but also how humans perceive the AI's role within the team. Studies show that fluid interaction patterns contribute to trust calibration, reduce cognitive load, and facilitate error recovery in dynamic contexts. Cognitive augmentation enhances decision-making by providing predictive insights, pattern recognition, and decision support. However, augmentation must be carefully aligned with human goals and workflows to avoid overreliance or disengagement. Trust in AI support hinges on its transparency, relevance, and adaptability to human intent. The role of GenAI is emerging. GenAI demonstrates clear benefits in early-stage design, brainstorming, and content generation. However, its outputs often lack traceability and explainability, introducing new challenges in accountability and ethical oversight. While promising, GenAI integration into real-world Human-AI teams remains underexplored and requires



empirical validation. This review contributes to the field by: (1) framing HATs as distributed cognitive systems rather than task-based pairings, (2) mapping the interactions among enablers that shape hybrid team effectiveness, and (3) identifying critical research gaps, particularly around trust calibration, augmentation alignment, and long-term collaboration. Methodologically, it also illustrates how Human-AI teaming can enhance the research process itself, through a semi-automated review pipeline supported by NLP, LLMs, and human oversight. The findings support ongoing developments in HCAI and align with the Industry 5.0 vision of synergistic, sustainable, and responsible collaboration between humans and machines. As complexity continues to define modern socio-technical systems, future research should prioritize longitudinal studies, cross-cultural validation, and domain-specific adaptation to ensure that Human-AI collaboration remains effective, ethical, and resilient across diverse contexts.

## Appendix A: Search Queries by Research Question

1. **RQ1: How is task allocation performed in Human-AI teams?** ("human-AI team" OR "Human-AI Teams") AND ("Task allocation" OR "Task distribution" OR "Task assignment"); ("human-AI team" OR "Human-AI Teams") AND ("task performance" OR "team performance" OR "task completion") AND ("allocation" OR "assignment")
2. **RQ2: What communication strategies are used in Human-AI collaboration?** ("human-AI team" OR "Human-AI Teams") AND ("communication strategies" OR "interaction strategies" OR "collaboration"); ("human-AI team" OR "Human-AI Teams") AND ("collaborative communication" OR "effective communication" OR "interaction patterns")
3. **RQ3: How does AI augmentation influence decision-making in teams?** ("human-AI team" OR "Human-AI Teams") AND ("AI augmentation" OR "augmentation" OR "AI-assisted decision making" OR "decision support"); ("human-AI team" OR "Human-AI Teams") AND ("AI impact" OR "AI influence" OR "human-augmented decision making")
4. **RQ4: What role does GenAI play in Human-AI teams?** ("human-AI team" OR "Human-AI Teams") AND ("Generative AI" OR "GenAI" OR "AI-powered tools"); ("human-AI team" OR "Human-AI Teams") AND ("trust" OR "transparency" OR "adaptability") AND ("Generative AI" OR "GenAI")

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